rabia_naz at StanceEval-2026: An Ensemble of Fine-Tuned Open LLMs for Arabic Stance Detection on Seen Targets

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Abstract

This paper presents a system for StanceEval-2026 Track 1, Arabic stance detection on seen targets. Systems in this track are trained on the provided targets and evaluated on a related target within the same thematic domain. Open large language models are fine-tuned with QLoRA on the original Mawqif training data and adapted to the test target through transductive self-training, in which unlabeled test tweets are added to training only when independent models agree on their label. The final prediction is a majority vote over three fine-tuned models, Qwen2.5-7B, the Arabic-native ALLaM-7B, and a sarcasm-aware variant that predicts the official sarcasm label before the stance label. The system scored 0.7564 Favg2 and placed 12th of 21 teams on the official leaderboard. Across seventeen scored variants, task fine-tuning outperformed zero-shot prompting at every model scale, heterogeneous fine-tuned models converged on 98% of test predictions, and further gains required decorrelated ensemble members. The sarcasm annotation supplied this decorrelation and improved the vote, while the sentiment annotation degraded it.

1 Introduction

Stance detection classifies a text as supportive, opposed, or neutral toward a given target. The task differs from sentiment analysis because a tweet with negative tone can still support its target. StanceEval-2026 (Albalawi et al., 2026a) evaluates Arabic stance detection on social media. Track 1 trains on three targets from the original Mawqif dataset and tests on a related target absent from the training data. Systems that memorize target-specific vocabulary therefore lose their advantage on the test set.

Three decisions define the submitted system. First, open instruction-tuned language models are fine-tuned on the official data with QLoRA. This configuration scored higher than all tested encoder ensembles and zero-shot models, up to 32B parameters. Second, transductive self-training adapts the fine-tuned models to the test target. A test tweet joins the training data only when independent models agree on its label, so the filter admits fewer but more reliable pseudo-labels. Third, the final vote combines members selected for decorrelation instead of individual strength. An Arabic-native backbone and a sarcasm-aware model each added 0.3 points through the vote.

The system scored 0.7564 Favg2 on the official test set and placed 12th of 21 participating teams on the official leaderboard. All experiments ran on a single Tesla T4 GPU on Kaggle and used no data beyond the official release. Negative results receive the same attention as positive ones because they delimit the method. Larger models did not improve accuracy, either zero-shot or fine-tuned. Combinations of the fine-tuned model with encoder ensembles reduced performance. The sentiment label, unlike the sarcasm label, carried no usable signal. The code is publicly available.[1]

2 Background

Track 1 provides an Arabic tweet and a target as input. The output is one of three stance labels, Favor, Against, or None. The official training material is the original Mawqif dataset (Alturayeif et al., 2022), with 3,502 training tweets and 619 development tweets across the targets COVID-19 vaccine, digital transformation, and women empowerment. Each tweet additionally carries a sentiment label and a sarcasm label. The task allows both as auxiliary supervision. The stance distribution is imbalanced, with 61% Favor, 29% Against, and 10% None in training. Tweets without a stance annotation are assigned the None class during prepro-

[1] https://github.com/iamRabia-N/StanceEval-2026

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cessing. The hidden test set is the Women Driving subset of the Mawqif-XT extension (Albalawi et al., 2026b). It contains 352 tweets on a target that is absent from training and related to the women empowerment target. Lexical overlap between the two topics is partial, and support for one does not imply support for the other. The official metric Favg2 is the macro F1 over the Favor and Against classes. The secondary metric Favg3 additionally includes None.

3 Related Work

SemEval-2016 formalized stance detection on Twitter and fixed the Favor, Against, and None scheme with macro-F1 evaluation (Mohammad et al., 2016). Arabic work then adopted fine-tuned encoders. AraBERT (Antoun et al., 2020) and MARBERT (Abdul-Mageed et al., 2021) led the first StanceEval edition (Alturayeif et al., 2024). TAO (Melhem et al., 2024) is a representative single-task AraBERT system from that edition. These systems train and evaluate on the same targets. Encoders that learn target-specific features have no mechanism to transfer them to a new target. The present system responds to this gap. An instruction-tuned language model fine-tuned with parameter-efficient methods (Hu et al., 2022; Dettmers et al., 2023) should generalize across targets better than encoders. The experiments confirm this over encoder ensembles and zero-shot models up to 32B parameters. The tested backbones include the Arabic-centric ALLaM (Bari et al., 2024) and JAIS (Sengupta et al., 2023).

The Mawqif dataset was annotated with sentiment and sarcasm labels to enable multi-task learning for stance (Alturayeif et al., 2022). Participants in the 2024 edition largely trained single-task systems and left the value of these labels untested (Melhem et al., 2024). This work measures that value at the language-model level with a controlled pair of experiments. One member is trained to predict sarcasm before stance and one is trained identically with sentiment. The results show the two signals are not interchangeable. The adaptation method builds on classical self-training (Yarowsky, 1995; Xie et al., 2020) and on the co-training principle that agreement between diverse views produces reliable labels (Blum and Mitchell, 1998). The agreement filter in this work applies that principle to heterogeneous fine-tuned models.

Table 1: Fine-tuning configuration shared by all ensemble members.

Parameter	Value
Adapter	LoRA, r=16, α=32
Target modules	q, k, v, o projections
Dropout	0.05
Quantization	4-bit NF4
Learning rate	2e-4
Warmup ratio	0.05
Epochs	2 (3 for sarcasm-aware)
Effective batch size	16
Sequence length	320–340
Optimizer	paged AdamW 8-bit
Seed	42
Hardware	1$\times$ Tesla T4 (16 GB)
Training time	3–4 h per member

4 System Overview

The final system is a majority vote over three QLoRA fine-tuned models. All members share the same supervision format. The model receives an instruction that names the target, the tweet text, and a request to answer with the label. The supervised answer is the label followed by the end-of-sequence token, so the training loss concentrates on the decision tokens instead of the tweet text. Table 1 lists the shared configuration.

Qwen2.5-7B-Instruct (Qwen Team, 2024) and ALLaM-7B-Instruct (Bari et al., 2024) are fine-tuned on the official tweets extended with self-training pseudo-labels. The Arabic-native backbone is included for decorrelation. The two models agree on 98.3% of test tweets and the remaining disagreements carry signal. A sarcasm-aware variant of Qwen2.5-7B completes the ensemble. It is trained to answer in the format `Sarcasm: Yes/No. Stance: X` with the official sarcasm labels. The format forces an irony decision before the stance decision on a topic where sarcastic support and sarcastic opposition are both frequent. This member is weak in isolation (0.6987). A sufficient share of its disagreements with the other members are corrections that raise the ensemble score.

Self-training adds pseudo-labeled test tweets to the training data of two members. The procedure has three steps. First, two or more previously trained systems label the 352 test tweets. Second, a tweet is accepted only if these systems agree on its label and the label is not None. Third, the accepted tweets are added twice to the 4,121 official tweets,

and the member is trained from its base model on this extended set. The procedure is applied once per member. For the Qwen2.5-7B member, the labeling systems are the first fine-tuned Qwen2.5-7B (0.7482 in Table 2) and the five-model encoder ensemble (the MARBERT+AraBERT ensemble in Table 2). They agree on 245 tweets, and the retrained member scores 0.7504. For the ALLaM-7B member, the labeling systems are the first fine-tuned Qwen2.5-7B, the self-trained Qwen2.5-7B, and the fine-tuned Qwen2.5-14B, and a tweet is accepted when at least two of the three agree. They agree on 346 tweets. The sarcasm-aware member is trained on the official data only. Both self-trained members enter the final vote. The acceptance criterion has a larger effect on accuracy than the number of pseudo-labels. A Qwen2.5-7B model trained with the same configuration on the 346 tweets of the three-model vote scored 0.7474, and a second self-training round with a lower confidence threshold on the encoder ensemble scored 0.6882 against 0.6984 for a single round, so no member receives more than one round.

The vote averages per-class scores. The two Qwen-based members contribute one-hot votes and ALLaM contributes its label probabilities. A label selected by both one-hot members always obtains the highest average score. When the one-hot members disagree, the ALLaM probabilities determine the final label. A parse-failure fallback assigns Favor. It did not activate in any reported run.

5 Experimental Setup

All models trained on Kaggle sessions with a Tesla T4 GPU. The software stack consists of PyTorch, Transformers, PEFT, and bitsandbytes. The 14B experiments additionally used Unsloth and llama.cpp for model loading. The library versions were PyTorch 2.10, Transformers 5.0, and PEFT 0.19, with bitsandbytes installed at run time in each session. Preprocessing normalizes alef and yaa variants, strips diacritics and elongations, replaces URLs and mentions with placeholder tokens, and splits hashtags because hashtag segments often carry stance information. Tweets are truncated to 300 characters inside the prompt. Prompts were written in English while the tweet text remained in Arabic. The full prompt texts appear in Appendix A. The Qwen members use greedy decoding with at most 4 new tokens and the sarcasm-aware member uses at most 12. Label scoring by first-token log-probability produced predictions identical to greedy decoding, so the fine-tuned models are confidently calibrated on this task. The ALLaM member is scored by first-token log-probability directly.

During the development phase, the development set validated the procedure on seen targets. The strongest configuration scored 0.8486 Favg2 on the official scorer. It exceeded all four organizer baselines, among them zero-shot Qwen2.5-72B at 0.8425 and JAIS-70B at 0.8337. During the evaluation phase, the development set joined the training data because no labeled data exists for the test target. The gap between development and test scores measures the difficulty of transfer to the test target. System selection on the test target used the official scorer with one controlled change per submission. Total compute across all reported experiments was approximately 60 GPU hours.

6 Results

Table 2 reports every major scored variant with its outcome and includes the approaches that failed. Fine-tuning produced the largest gain. The QLoRA Qwen2.5-7B model scored 0.7482 and no encoder ensemble or zero-shot model matched it. The 3.4-point improvement over the best prompting-based system is the largest single gain across all experiments. Model scale saturated for both zero-shot and fine-tuned models. Zero-shot accuracy remained near 0.70 across the 7B, 14B, and 32B models. A fine-tuned 14B scored below the fine-tuned 7B, so compute on this task is better spent on adaptation than on parameters. Ensemble diversity helped only when the added member was informative. The Arabic-native member and the sarcasm-aware member each added 0.3 points through the vote. A sentiment-aware member built identically reduced the vote by 0.3. Combinations of the fine-tuned model with the earlier encoder ensembles also reduced accuracy. The fine-tuned model had already absorbed the knowledge of the encoders. Curriculum ordering of official and pseudo data with NEFTune noise produced no change (0.7504).

Three Qwen2.5-7B runs with the same configuration and different pseudo-label sets scored 0.7474, 0.7482, and 0.7504. This range of 0.3 points on 352 test items bounds the effect of small changes. Differences of similar size in the table are therefore not conclusive. The submitted system scored 19 points above the official test baseline (0.5643). The secondary Favg3 is 0.5233. This value is low

Table 2: Official leaderboard submissions on the test set, in chronological order. Rows that begin with + add the named component to the system in the row above. No post-deadline results are reported.

System	Favg2	Outcome
MARBERT+AraBERT ens.	0.6769	baseline
+ zero-shot LLM fusion	0.6788	no gain
+ self-training	0.6984	+2.2
+ 3-prompt voting	0.7138	+1.5
Zero-shot 14B / 32B	0.70 / 0.70	saturates
QLoRA Qwen2.5-7B	0.7482	+3.4
+ self-training	0.7504	+0.2
QLoRA + encoder blends	0.70–0.72	reduced
QLoRA Qwen2.5-14B	0.7325	below 7B
Vote: Qwen + ALLaM	0.7534	+0.3
Sarcasm-aware member	0.6987	weak individually
Vote: + sarcasm-aware	0.7564	submitted
Sentiment-aware in votes	0.7456	reduced

because the agreement filter excludes None from the pseudo-labels, so the final members rarely predict the None class. The system dropped 9 points from 0.8486 on the seen targets to 0.7564 on the test target. This drop quantifies the difficulty of target transfer for the method. Four fine-tuned models with different backbones and pretraining data agreed on at least 98% of test predictions. The official training data therefore appears to determine the predictions of these fine-tuned models. Only members with distinct error profiles moved the score.

The error analysis focused on the sarcasm-aware member. It disagreed with both other members on 71 tweets, 20% of the test set, and predicted sarcasm on 74 of the 352 test tweets. Its most visible deviation is the assignment of None to 26 tweets where each other member assigns None once. These None votes remove tweets from its Favor and Against recall and lower its individual score. Its corrections occur mainly on sarcastic tweets in which the literal wording contradicts the intended stance. The most frequent pattern is a supportive tweet posted under the opposition slogan hashtag (you will not drive). One such tweet, translated from Arabic, states that the author already drives far better than men. Another announces that the author will soon carry her license in her bag. Both other members follow the slogan and assign Against, while the sarcasm-aware member assigns Favor. The reverse pattern also appears. One tweet describes the Saudi woman as a queen who gave up her crown on the day she started to serve everyone by driving. The wording is positive toward women, the intended stance opposes driving, and only the sarcasm-aware member assigns Against. The sentiment-aware member showed a different pattern. Its disagreements followed the tone of the tweet. They carried no information about the stance toward the target, so its votes repeated existing errors. This result confirms the distinction between stance and sentiment on real data. It also explains why the sarcasm signal helps the vote while the sentiment signal does not.

7 Conclusion

A QLoRA fine-tuned open language model scored 0.7564 Favg2 on StanceEval-2026 Track 1, Arabic stance detection on seen targets. The model was adapted through transductive self-training and combined by majority vote with an Arabic-native member and a sarcasm-aware member. Fine-tuning was stronger than zero-shot prompting at all tested scales. Larger models added no accuracy. Heterogeneous fine-tuned models converged on the same predictions. Decorrelated members provided the gains beyond the convergence point. The official sarcasm annotation was the informative auxiliary signal. Future work includes a member that combines the Arabic-native backbone with sarcasm supervision, improved pseudo-label selection, and a study of the convergence limit across additional targets.

Limitations

The transductive self-training adapts the members to the specific test set, so every new target requires a new round of pseudo-labeling and retraining. Hyperparameters transferred from seen-target validation without test-side tuning. The agreement filter lowers None recall and therefore Favg3. The evaluation covers a single test target, so the generality of the conclusions across other targets is untested.

Ethical Considerations

The system classifies stance in public Arabic tweets and is intended for aggregate analysis of public discourse. It is not intended for conclusions about individual users. The Mawqif tweets cover Saudi public debates, so trained models match the topical and dialectal coverage of this sample. No personal data was collected beyond the official dataset. User mentions were replaced with placeholder tokens during preprocessing.

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A Prompts

Both prompts are wrapped in the model's chat template, and the tweet is truncated to 300 characters.

Prompt used by the stance members:

```
Classify the stance of this Arabic tweet
toward the target "TARGET". Answer with
exactly one word: Favor, Against, or
None.
Tweet: TWEET
Answer:
```

Prompt used by the sarcasm-aware member:

```
Analyze this Arabic tweet about the target "TARGET". First decide if the tweet is sarcastic, then decide the stance of the author toward the target. Answer in exactly this format: Sarcasm: Yes or No. Stance: Favor or Against or None.
Tweet: TWEET
Answer:
```

Model checkpoints:

```
Qwen/Qwen2.5-7B-Instruct
ALLaM-AI/ALLaM-7B-Instruct-preview
```